#Bayes' theorem, #Probability, #Statistics

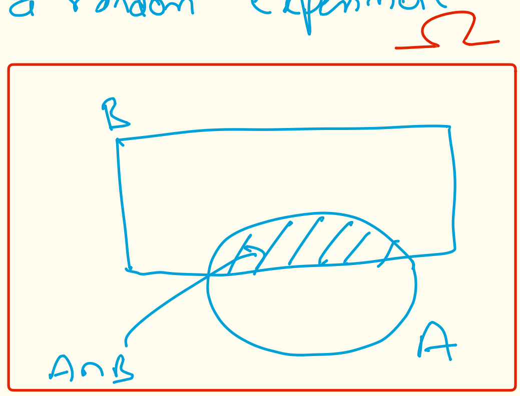
Notebook

Day - 4

Bayes' Theorem.

Bayes' Theorem.

A, B are two events (outcomes) of a random experiment.



$$P(A|B) = \frac{P(A\cap B)}{P(B)}$$

$$P(A\cap B) = P(A|B)\,P(B)$$

$P(B|A) = ?$ (given that I have a +ve test result prob. that I have cancer).

$$\underbrace{P(B|A)}_{\text{Posterior (updated prior after the info 'A')}} = \frac{\overbrace{P(A|B)}^{\text{update (likelihood)}} \cdot \overbrace{P(B)}^{\text{prior (initial)}}}{P(A) \rightarrow \text{evidence or normalizing constant}}$$

$$P(B|A) = \frac{P(A\cap B)}{P(A)}$$

$$P(A|B) = \frac{P(A\cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A)}$$

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A|B)\cdot P(B) + P(A|B^c)\cdot P(B^c)}$$

[Law of total probability]

$$P(B_j|A) = \frac{P(A|B_j)\,P(B_j)}{\sum_{j=1}^{n} P(A|B_j)\,P(B_j)}$$

[definition of total prob.]

Ex 1:

Test is 99% accurate. but the disease is very rare. (0.001 of population have this disease)

A = positive test +

B = disease

$$P(\text{disease} \mid +) = \frac{P(+ \mid \text{disease}) \cdot P(\text{disease})}{P(+)}$$

$$= \frac{P(+ \mid \text{disease})\, P(\text{disease})}{P(+ \mid \text{disease})\, P(\text{disease}) + P(+ \mid \text{no disease})\, P(\text{no disease})}$$

$$= \frac{0.99 \times 0.001}{(0.95 \times 0.001) + (0.01 \times 0.999)}$$

$$\approx 0.09.$$

Ex 2: Drug Testing.

Test is 90% accurate.

and 10% user rate.

B = do you actually use.

A = test score

$$P(+ \mid \text{user}) = 0.9$$

$$P(- \mid \text{non-user}) = 0.9$$

$$P(\text{user} \mid +) = ?$$

Using Bayes' theorem:

$$P(\text{user} \mid +) = \frac{P(+ \mid \text{user}) \cdot P(\text{user})}{P(+)}$$

$$= \frac{P(+ \mid \text{user}) \cdot P(\text{user})}{P(+ \mid \text{user}) \cdot P(\text{user}) + P(+ \mid \text{non-user}) \cdot P(\text{non-user})}$$

$$= \frac{0.9 \times 0.1}{0.9 \times 0.1 + 0.1 \times 0.9}$$

$$= 0.5$$

Ex 3:

how sensitive it is to the thing we are trying to detect.

$p(+ \mid \text{user}) = 0.90$ [Sensitivity]

$p(- \mid \text{non-user}) = 0.80$ [Specificity].

measure of how many false positives we're going to get in non-user population.

$p(+ \mid \text{non-user}) = 0.20$

$$p(\text{user} \mid +) = \frac{p(+ \mid \text{user}) \cdot p(\text{user})}{p(+ \mid \text{user}) \cdot p(\text{user}) + p(+ \mid \text{n-user}) \cdot p(\text{n-user})}$$

$$= \frac{(0.9) \times (0.1)}{(0.9 \times 0.1) + (0.2)(0.9)} = \frac{1}{3}$$

Independence.

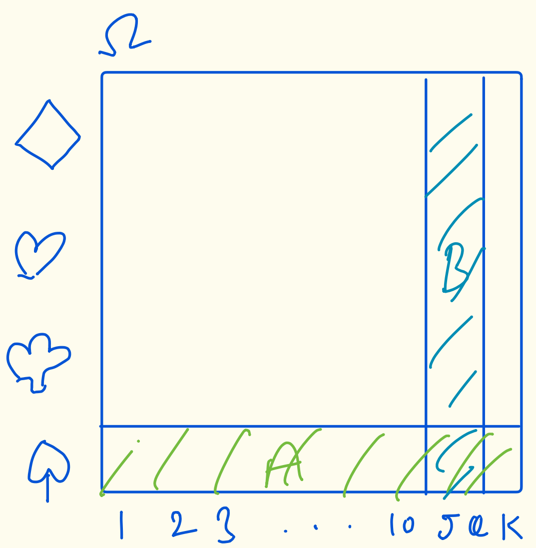
Two events A and B are independent if knowing one occurred gives no extra info about the other.

$$P(A|B) = P(A)$$
$$P(B|A) = P(B)$$
$$\boxed{P(A\cap B) = P(A)\,P(B)}$$

IID = Independent and Identically Distributed.

$$P(A) = P(A|B) = \frac{P(A\cap B)}{P(B)}$$

$$\boxed{\therefore\ P(A\cap B) = P(A)\cdot P(B)}$$



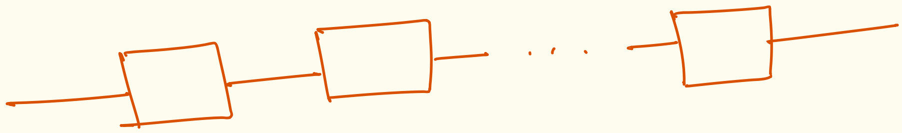
A = Spade $\quad P(A) = \frac{1}{4}$

B = Q $\quad P(B) = \frac{1}{13}$

$$P(A|B) = P(A)$$
$$P(B|A) = P(B)$$
$$P(A\cap B) = P(A)\,P(B) = \frac{1}{52}.$$

Reliability

n components in series.



prob(single element failure) $= p = 0.05$

prob(series failure) = hard to compute.

$= 1 - p$(no any failing)

$= 1 - (1-p)^n$

If $n = 10$, $p = 0.05$

no any failing $= 0.6$

(Series makes things less reliable).

For Parallel.

$p(\text{failure}) = p^n = 10^{-13}$.